

Noriyuki (Noly) Higashide

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Interests

Summary	I explore how metascience and AI can turn scientific knowledge into innovation, entrepreneurship, and real-world societal impact.
Keywords	Science of Science, AI for Science, Computational Social Science, Network Science, Technology Management, Knowledge Translation, Startup Ecosystem, Translational Research

Education

2020.09 - 2025.09	The University of Tokyo , Tokyo, Japan Ph.D. in Technology Management for Innovation, Graduate School of Engineering Advisor: Prof. Ichiro Sakata, GPA: 3.9/4.0, with Department Chair's Award for my dissertation
2013.04 - 2016.03	The University of Tokyo , Tokyo, Japan M.Sc. in Electrical and Electronic Engineering, Graduate School of Engineering Advisor: Prof. Yuichiro K. Kato, GPA: 3.4/4.0
2011.04 - 2013.03	Chiba University , Chiba, Japan B.Eng. in Electrical and Electronic Engineering, School of Engineering, GPA: 3.6/4.0
2006.04 - 2011.03	Hakodate National College of Technology , Hokkaido, Japan A.Eng. in Electrical and Electronic Engineering, GPA: 3.7/4.0

Experience

2026.04 -	Impact Fellow , Northwestern Innovation Institute, Kellogg School of Management, Northwestern University Evanston, Illinois, United States.
2026.04 -	Postdoctoral Research Fellow , Center for Science of Science and Innovation, Kellogg School of Management, Northwestern University Evanston, Illinois, United States.
2024.10 -	Founder / CEO , fukan Inc. Tokyo, Japan. • Building an AI tool that identifies optimal collaboration partners for innovation activities. • Original CV wording: Run the firm that develops specialized AI solutions that extract scientific and technological information from extensive literature databases, streamlining corporate innovation and R&D investment decisions.
2020.10 - 2024.09	Senior Data Scientist / Project Manager , SRE Holdings Corporation Tokyo, Japan. • In Sony-funded post-IPO startup, directed and taught hands-on AI training programs for executives at leading financial and pharmaceutical companies. • Drove new business development in climate tech, formulated over 10 consulting proposals, and conducted team-building activities. • Identified investor behavior trends in major brokerage firms by network analysis and machine learning with a self-developed data handling system on cloud for terabyte-scale financial data.

2019.07 - 2024.09	Consultant / Data Scientist , Freelance Tokyo, Japan. • Suggested digital transformation direction to a major insurance company based on insights from sales and call data analysis. • Advised on the construction of a chat system utilizing generative AI. • Designed IoT system architecture composed of cameras, edge devices, and video analysis cloud platforms.
2016.10 - 2019.06	Consultant / Data Scientist , IBM Japan Ltd. Tokyo, Japan. • Led 6+ AI projects as a project manager and technical lead for major enterprise clients, specializing in chat system, speech recognition systems, and text data analysis. • Co-presented a project achievement as advanced case study on speech recognition AI with a client at an internal global conference IBM THINK 2019. • Contributed to community building by organizing internal hackathons.
2015.07 - 2016.09	Research Assistant Intern , Sony Computer Science Laboratories Inc. Tokyo, Japan. • Supported to test a novel 3D lighting device; implemented electric circuits.

List of Research Products

Journal Articles

(‡: Corresponding author)

- [1] N. Higashide‡, K. Nishimoto, K. Asatani and I. Sakata, Asymmetric returns to basic-applied career shifts. *Quantitative Science Studies* (2026).
- [2] N. Higashide‡, T. Miura, Y. Tomokiyo, K. Asatani and I. Sakata, Mid-career pitfall of consecutive success in science. *Scientific Reports* 14, 28172 (2024).
- [3] N. Higashide‡, Y. Zhang, K. Asatani, T. Miura and I. Sakata, Quantifying advances from basic research to applied research in material science. *Technovation* 135, 103050 (2023).
- [4] N. Higashide, M. Yoshida, T. Uda, A. Ishii and Y. K. Kato, Cold exciton electroluminescence from air-suspended carbon nanotube split-gate devices. *Applied Physical Letter* 110, 191101 (2017).

Conference Papers

- [5] R. Kaku, K. Asatani, N. Higashide, I. Sakata, Theory-to-application transition in mathematics and computer science studies. 2024 Portland International Conference on Management Engineering and Technology (2024).
- [6] N. Higashide†, T. Miura† and I. Sakata, Different hot streaks for twin peaks in academic careers. Proceedings of International Conference of the International Society for Scientometrics and Informetrics 2023 (2023). (†equal contribution)
- [7] N. Higashide, K. Asatani and I. Sakata, Domestically interacted scientists drive nanocarbon applied research in China. 2023 Portland International Conference on Management Engineering and Technology (2023).
- [8] N. Higashide, K. Asatani and I. Sakata, Identifying quantitative research levels in nanocarbons using semi-automatically extracted vocabularies. 2022 Portland International Conference on Management Engineering and Technology (2022).

Conference Presentations

- [9] N. Higashide, H. Miyamoto, C. Mejia, I. Sakata, Y. Kajikawa, Academic Knowledge Accumulation Drives Venture Capital Investments. The 4th International Conference on Science of Science and Innovation (2025).
- [10] N. Higashide, K. Nishimoto, K. Asatani, I. Sakata, Basic-to-Applied Transitions Boost Scientific Impact. The 4th International Conference on Science of Science and Innovation (2025).
- [11] Y. Tomokiyo, T. Miura, N. Higashide, K. Asatani and I. Sakata, Quantitative evaluation of the degree of consecutive success extraction of the hot-streak model. The 38th Annual Conference of the Japanese Society for Artificial Intelligence (2024).
- [12] N. Higashide, H. Miyamoto, C. Mejia, I. Sakata and Y. Kajikawa, Understanding Venture Capital Investment Trends through Social and Academic Recognition. The 3rd International Conference on Science of Science and Innovation (2024).
- [13] N. Higashide, H. Miyamoto, C. Mejia and Y. Kajikawa, Diffusions and Interactions of Technological Trends in Academic Publications, Venture Capital Investment, and News. The 1st Meeting on Science of Science in

Japan (2024).

- [14] N. Higashide†, T. Miura†, K. Asatani and I. Sakata, Hot Streaks Occur in Early and Late Career Stages Beyond Random. The 2nd International Conference on Science of Science and Innovation (2023). (†equal contribution)
- [15] N. Higashide, T. Uda, M. Yoshida, A. Ishii, Y. K. Kato, Electroluminescence from individual air-suspended carbon nanotubes within split-gate structures. APS March Meeting 2016, 61 Number 2 (2016).

Working Paper

- [16] T. Nagano, N. Higashide, G. Kurata, R. Tachibana, T. Nakayama, H. Shirouzu, Trial of junior high-School student speech recognition in collaborative learning, knowledge constructive Jigsaw method. Information Processing Society of Japan Technical Report, Spoken Language Processing 124 (2018).

Media / Press

- [17] Hot Streak - “Exploration and Exploitation” to create a series of hits. UTokyo Integrated Report 2023 - IR Cubed -, p78-79 (Nov. 2023).
- [18] Close to a Nobel Prize? Toshihide Amari’s AI Research as the “Sleeping Beauty”. Nikkei (Dec. 1, 2024).

Working in Progress

- [19] Technological hypes and venture capital investment: artificial intelligence, biotechnology, and cleantech. (Co-author; with H. Miyamoto and Y. Kajikawa).
- [20] Discovering startup-ready scientists from co-author networks. (Co-author).

Academic Presentations

2025.06	Poster. The 4th International Conference on Science of Science and Innovation, Copenhagen, Denmark.
2024.09	Invited Seminar. The 85th JSAP Autumn Meeting 2024, Niigata, Japan.
2024.08	Invited Presentation. MIT Quantum Measurement Group, Cambridge, MA, USA (via zoom).
2024.07	Poster. The 3rd International Conference on Science of Science and Innovation, Washington, DC, USA.
2024.03	Poster. The 1st Meeting on Science of Science, Tokyo, Japan.
2023.07	Talk. 2023 Portland International Conference on Management Engineering and Technology, Monterrey, Mexico.
2023.07	Talk. International Conference of the ISSI 2023, Bloomington, IN, USA.
2023.06	Poster. The 2nd International Conference on Science of Science and Innovation, Evanston, IL, USA.
2022.08	Talk. 2022 Portland International Conference on Management Engineering and Technology, Portland, OR, USA.
2016.03	Talk. APS March Meeting 2016, Baltimore, MD, USA.
2015.02	Poster. The 48th FNTG General Symposium, Tokyo, Japan.

Academic Services and Community Projects

Academic Services

2026.04 -	Journal Reviewer , <i>Quantitative Science Studies</i>
2023.10 -	Journal Reviewer , <i>Scientometrics</i>
2023.01 -	Journal Reviewer , <i>Technological Forecasting and Social Change</i>
2023.01 - 2023.03	Program Manager , Deep Innovation Creation Ecosystem (DICE), School of Engineering, The University of Tokyo Coordinated educational tours to key venture capital and pharmaceutical sites in Boston and New York to understand the startup ecosystems in the U.S.
2022.04 - 2022.09	Teaching Assistant , Deeptech Entrepreneurship Incubation Program, School of Engineering, The University of Tokyo Guided participants in strategic planning for Deeptech startups with a long-term vision for 2050.

2014.04 - 2016.09 **Technical Assistant**, Institute of Engineering Innovation, The University of Tokyo
Implemented automated electrical and optical measurement systems for nano-scale home-built fabricated devices.

Community Projects

2018.10 - 2026.09 **Technical Director / Board**, Approved Specified Nonprofit Corporation CALL4
Provided technical oversight in litigation funding online web platform.

2017.04 - 2017.09 **Program Organizer**, Watson Hackathon, IBM Japan Ltd.
Hosted an enterprise-wide hackathon event to enhance understanding of AI solutions across business functions.

2015.04 - 2016.09 **Program Chair**, Student Organization TEDxUTokyo
Led to organize a cross-disciplinary, university-level conference with 500+ attendees and 16 speakers.

Awards, Honors and Scholarships

Awards

2026.03 **Department Chair's Award** for my dissertation "Decoding Scientific Career Paths through Hot Streaks and Basic-Applied Research".

2023.06 **Winner** in Knowledge Track, ICSSI Metascience Hackathon.

2022.06 **CEO Award**, SRE Holdings Corporation. (Top 2%)

2022.02 **Best Business Plan**, Deeptech Entrepreneurship Incubation Program. (Top 2)

2020.03 **Finalist** in New Venture Challenge, The University of Chicago.

Honors

2018.05 **Selected Member** for an exclusive tour to prestigious institutions in Singapore for recognizing professional contributions, IBM Japan Ltd. (Top 1%)

2011.03 **Special Commendation** for successfully organizing an alternative event for the cancelled school festival due to an infectious disease, HNCT. (Top 0.1%)

Scholarships

2013.04 - 2015.03 Scholarship for Graduate Student, THE USHIO FOUNDATION. (Top 5%)

2012.04 - 2015.03 Kazuko Inoue Scholarship, Ningenjuku Foundation. (Top 10%)

Skills

Programming	Python, MATLAB, SQL, JavaScript, Node.js, Java
Computing	Unix, LaTeX, OriginLab, MS Office, Markdown
Language	English (fluent), Japanese (native)

Research Projects

2023.05 - 2023.10 **AI and Flexible Electronics for Healthcare Business**, The University of Tokyo
Executed development of a machine learning algorithm that estimates blood pressure from pulse waves measured by flexible organic photodetectors and ideation innovative business applications utilizing it. Advisor: Prof. Takao Someya and Prof. Yutaka Matsuo.

2023.09 - 2023.10 **Summer School for Large Language Models**, The University of Tokyo
Completed advanced training in generative AI and Large Language Models, implementing enhancements for NLP task precision. Advisor: Prof. Yutaka Matsuo.

2017.10 - 2019.07 **Speech Recognition in Education**, IBM Japan Ltd.
Developed speech recognition applications with AI using Node.js and Java; co-presented with a client on the practices of speech recognition in education at IBM Think 2019. Advisor: Ph.D. Ryuki Tachibana, Ph.D. Tohru Nagano and Prof. Hajime Shirouzu.

2016.07 - 2017.06	Across , Personal Project Enhanced readability of the arXiv papers with smartphone reader app.
2015.07 - 2016.09	AnyLight , Sony Computer Science Laboratories Inc. Engineered a next-generation lighting device, specializing in the design of optical lens arrays and the execution of electrical circuitry. Advisor: Ph.D. Yuichiro Takeuchi.
2012.04 - 2013.03	Undergraduate Thesis , Chiba University Executed simulations of electron and phonon spatial distribution by Fortran.
2010.04 - 2011.03	Graduation Thesis , Hakodate National College of Technology Created a simulation model to assess CO2 emission reduction in future power grid systems.

Business Projects

2023.08 - 2024.09	Testing business hypotheses based on research outcomes , Concretized a business hypothesis, that support innovation investment using techniques of the doctoral research, through hearings with leading venture capitals and major corporations.
2021.10 - 2022.03	Deeptech Entrepreneurship Incubation Program , Crafted a startup business plan for decarbonization using advanced technology from The University of Tokyo, winning second place in a competition judged by experts including the university president and corporate executives.
2019.10 - 2020.03	Smartmil: business planning with Chicago University MBA , Developed a SaaS business plan for dental clinic management, which won finalists in an accelerator programs New Venture Challenge.
2019.03	Quick AI prototyping in Bay Area , Explored a startup idea using speech recognition AI with venture capitalists and engineers in Silicon Valley.
2017.04 - 2017.08	Business learning course in Keio University , Collaborated with diverse professionals across various industry and ages, resulting co-authored recommendations for social issues.

Selected Consulting Projects

2022.07 - 2022.12	Major Pharmaceutical Company , AI Training for Executives - Designed and delivered four 1.5-hour training sessions (video and in-person), including content creation, video recording/editing, and lectures. - Adjusted content flexibly based on feedback from participants, collaborating with counterparts. - Addressed individual questions in detail, improving executive understanding of data and AI usage, resulting in high customer satisfaction.
2021.01 - 2021.09	Major Securities Company , Python Data Analysis Training - Led the project from proposal to delivery, including planning, curriculum design, lecture material preparation, and participant evaluation. - Worked closely with the training office to ensure smooth progress. - Developed a skill assessment sheet to measure the training's impact, showing significant improvement in participants' data analysis skills, which satisfied the client.
2018.10 - 2019.03	HR Services Company , Job Seeker and Job Matching - Managed an agile development process, clarifying customer needs through prototyping. - Developed a matching method using deep learning and natural language processing to estimate word similarity for matching job postings with job seekers. - Supported the development of an AI-powered matching service with advanced technical insights, earning customer satisfaction.

- 2017.10 - 2018.06 **Major Content Company**, New Product Development
- Designed dialogue functions for a chatbot system.
 - Conducted user interviews for qualitative evaluation of dialogue quality.
 - Collaborated with clients and partner companies to define specifications and manage the development schedule.
 - Identified challenges, proposed solutions, and contributed to the planning of new service features.
- 2017.01 - 2017.12 **Major Manufacturer**, New Product Development
- Designed and managed development processes in collaboration with partner companies.
 - Facilitated client meetings, identifying issues and defining specifications.
 - Designed dialogue functions for IoT systems used in conversational robots.
 - Conducted qualitative evaluations of dialogue functions and provided technical support over a year, leading to the successful product release.